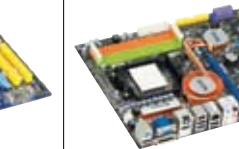


AMD Motherboards										
Brand Model No.	ASUS Crosshair II Formula	ASUS M3N-HT Deluxe	ASUS M3N78 PRO	Biostar GF8200 M2+		Biostar TA790 GX XE	Gigabyte MA78GPM-DS2H	Jetway HA06	Jetway HA07-LF	MSI DKA790GX Platinum
										
Chipset	NVIDIA nForce 780a SLI	NVIDIA nForce 780a SLI	NVIDIA GeForce 8300	NVIDIA GeForce 8200		AMD 790GX	AMD 780G	AMD 780G	AMD 790GX	AMD 790GX
Socket	AM2+	AM2+	AM2+	AM2+		AM2+	AM2+	AM2+	AM2+	AM2+
Northbridge	NVIDIA nForce 780a SLI	NVIDIA nForce 780a SLI	NVIDIA GeForce 8300	NVIDIA GeForce 8300		AMD 790GX	AMD 780G	AMD 780G	790GX	AMD 790GX
Southbridge						AMD SB750	AMD SB700	AMD SB700	AMD SB750	AMD SB750
Price (Rs)	15,450	15,450	6,150	3,900		6,600	8,900	5,500	6,750	9,600
+	Good bundle	Feature rich	Clean organized layout	Great value for money		Support for 16 GB of RAM	Good overall performance	Inexpensive CrossFire board	Great performer	Good performer
-	Expensive	Expensive	Slightly insufficient cooling	Poor cooling solution		Nothing in particular	Expensive	Heatsink lacks quality	Heatsink lacks quality	Slightly expensive
Grand Totals (Out Of 100)	69.45	63.20	61.61	52.55		60.12	61.55	61.34	64.18	63.97
Features (Out Of 80)	45.82	41.44	40.08	33.25		36.19	38.54	37.87	39.97	40.85
Performance (Out Of 20)	20.21	20.03	21.53	19.30		23.93	23.01	23.47	24.21	23.13
Features										
Memory Type / Max Freq. Supported	DDR2 / 1066 MHz	DDR2 / 1066 MHz	DDR2 / 1066 MHz	DDR2 / 1066 MHz		DDR2 / 1066 MHz	DDR2 / 1066 MHz	DDR2 / 800 MHz	DDR2 / 1066 MHz	DDR2 / 1066 MHz
Memory Supported / No. Of slots	8 GB / 4	8 GB / 4	8 GB / 4	8 GB / 4		16 GB / 4	16 GB / 4	8 GB / 4	8 GB / 4	8 GB / 4
Power Saving features (with GPU)	✓	✓	✓	✓		✗	✗	✗	✗	✗
No. Of SATA / PATA Ports	6 / 1	6 / 1	6 / 1	6 / 1		6 / 1	5 / 1	6 / 1	6 / 1	5 / 1
No. Of Rear E-SATA Ports	0	1	0	0		0	1	1	1	1
HDMI / DVI / VGA	✓ / ✗ / ✓	✓ / ✗ / ✓	✓ / ✗ / ✓	✗ / ✓ / ✓		✓ / ✓ / ✓	✓ / ✓ / ✓	✗ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓
Multi GPU Support (SLI / CrossFire / N)	✓ / SLI (3 way)	✓ / SLI (3 way)	✗	✗		✗	✗	✓ / CrossFire	✓ / CrossFire	✓ / CrossFire
Audio Controller / Channels / HD Compliant (Y / N)	ADI 1988B (SupremeFX) / 8 / Y	ADI 1988B / 8 / ✓	ALC1200 / 8 / Y	ALC662 / 6 / ✓		ALC888 / 8 / Y	ALC 889A / 8 / Y	ALC888 / 8 / Y	ALC888 / 8 / Y	ALC888 / 8 / ✓
S / PDIF / Optical Port	✓ / ✓	✓ / ✓	✓ / ✗	✓ / ✗		✗ / ✗	✗ / ✓	✗ / ✗	✗ / ✗	✗ / ✓
No Of LAN Ports / 10 / 100 / Gigabit	2 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit		1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit	1 / Gigabit
Integrated WiFi (Y / N)	✗	✗	✗	✗		✗	✗	✗	✗	✗
No. Of Rear USB / FireWire Ports	6 / 1	4 / 0	6 / 1	4 / 0		4 / 0	4 / 1	4 / 0	4 / 0	6 / 1
All Solid State Capacitors (Y / N)	✓	✓	✓	✗		✓	✓	✓	✓	✓
Overclocking Options										
CPU / FSB / Memory	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓		✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓
Overvolting (CPU / Memory / FSB / PCIe / Northbridge)	✓ / ✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✗ / ✓	✓ / ✓ / ✓ / ✗ / ✓		✓ / ✓ / ✓ / ✗ / ✓	✓ / ✓ / ✓ / ✗ / ✓	✓ / ✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✗ / ✓
Package Bundle										
No of SATA / PATA Cables Provided	3 / 1	2 / 1	3 / 1	1 / 1		2 / 1	2 / 1	4 / 1	4 / 1	2 / 1
Key bundled items	Company of Heroes - Opposing Fronts, SupremeFX, LED poster	Extra heatpipes kit								
Performance										
PC Mark 2005										
CPU	7653	7643	7612	7625		7563	7608	7598	7611	7621
Memory / Graphics	4518 / 1933	4954 / 1925	4906 / 2183	4411 / 1839		4466 / 2579	4686 / 2384	4938 / 2550	5069 / 2623	4422 / 2405
HDD	6125	6136	6121	6148		6016	6074	6067	6089	6068
Overall	5476	5336	5572	5286		5731	5646	5702	5756	5635
SiSoft Sandra 2008										
CPU Arithmetic (Dhrystone / Whetstone)	38963 / 33222	39010 / 33150	39007 / 33125	38990 / 33068		39343 / 33524	36618 / 33624	39338 / 33331	39348 / 33428	39369 / 33514
CPU Multimedia (Integer / Floating)	97833 / 129078	97222 / 129100	97844 / 128099	97864 / 129080		98749 / 130176	98670 / 130485	98632 / 130160	98657 / 130153	98662 / 130189
Drive Index (MBps) / Random Access Time (ms)	80 / 13	80 / 13	80 / 12	80 / 13		80 / 13	79 / 13	80 / 13	80 / 13	80 / 12
Memory Bandwidth Score (Integer / Floating)	5218 / 5220	5238 / 5192	5158 / 5165	5165 / 5167		5216 / 5213	5103 / 5101	5062 / 5064	5177 / 5173	5174 / 5176
3D Mark 2005:										
3D Marks / CPU Score	2232 / 7989	2248 / 8135	2554 / 8315	1819 / 7929		3649 / 5637	3501 / 8243	3545 / 8177	3676 / 8101	3299 / 7653
Video Encoding (DivX 6.2) Time	91	93	94	94		94	94	95	94	93
File Transfer - 4 GB File	200	200	201	197		201	198	200	200	198
Doom 3 (640x480 / 1024x768 - ultra no AA/AF)	50.4 / 26.7	50.5 / 26.6	60.5 / 32.9	46.6 / 25.0		60.2 / 35.6	57.2 / 33	57.7 / 33.8	60 / 35.4	52.6 / 30.7
Company of Heroes (800x600/1280x1024)	14.6 / 8.7	14.3 / 8.9	17.3 / 10.7	12.9 / 8.1		32.4 / 17.5	27.2 / 13.6	29.6 / 15.5	32.1 / 17.6	30 / 16.6

Features

If you want features, then the ASUS Crosshair II Formula comes with pretty much everything. It is a NVIDIA 780a board that supports 3-way SLI. It has two Gigabit LAN ports. There is no DVI port on the rear, but you will get HDMI and the usual VGA connector. The same is missing with the M3N-HT Deluxe. The Crosshair II Formula being another high-end ASUS board has loads and loads of features for overclockers.

Another equally priced board as the Crosshair II Formula is the ASUS M3N-HT Deluxe. This too runs off the NVIDIA 780a chipset but has a few missing features – mainly the second Gigabit LAN port, and also the SupremeFX audio card.

There is a thorough breakdown of voltages for each of the components and the user gets very fine control of each one of them. This is what you typically get from a true high-end ASUS board.

The only GeForce 8300 based motherboard from ASUS is the M3N78 PRO. Priced at less than half the price of the the other two ASUS boards, it has the same number of features except for some limited tweaking options in the BIOS for the PCI-E component. Like the GeForce 9300, the ZOTAC GeForce 8300 too has a few features to overclock the system.

The two Jetway boards the HA06 and the HA07 both have LED displays that also show the

temperatures which is a great feature. They also have on-board buttons for powering the board up. These motherboards are definitely worth looking at as they are both very cheap but come with a very good and detailed BIOS.

If you need a CrossFire supported board with unlimited features, then the MSI KA790GX Platinum is one to look out for. It comes with all the display connectivity options as well as Firewire, E-Sata and six USB ports.

Layout

The ASUS Crosshair II Formula is one of the more attractive motherboards out there. The mix of red copper and a blue coated southbridge heatsink. There’s also a lit-up Republic of Gamers badge right next to the north-bridge. Loads of copper pipings run all over the board. There are neat little lit buttons for powering up and also resetting the computer. A lit-up button on the rear of the motherboard allows users to reset the CMOS without

having to open up the cabinet and clear CMOS using a jumper or in some extreme caseses, even removing the battery by hand.

In comparison to the Crosshair II Formula and the M3N78 PRO, the M3N-HT Deluxe is a tiny bit more crowded. Once again, there are the similar lines of pipe running from up in the motherboard to the southbridge. We talked about how the M3N78 PRO from ASUS was cheaper and a little handicapped as compared to the other high-end end boards from ASUS. A few other